MILITARY SPECIFICATION

INDICATOR, HORIZONTAL SITUATION ID-1013(*)/A

This specification is approved for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers design and all performance requirements for the procurement of one type of course-bearing-heading-situation display indicator, which is a component of the integrated flight instrumentation system. The indicator displays course, bearing, heading, situation information and flight director information controlled by remote circuits.

1.2 Associated equipment. The course bearing heading situation indicator manufactured in accordance with this specification is a component of an integrated flight instrumentation system conforming to MIL-S-23535. The other components are the remote attitude director indicator manufactured in accordance with MIL-I-23524 and the flight director computer manufactured in accordance with MIL-C-23367.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 53), Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610

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SPECIFICATIONS

FEDERAL

BB-N-411 - Nitrogen, Technical.

QQ-C-320 - Chromium Plating (Electrodeposited).

QQ-N-290 - Nickel Plating (Electrodeposited).

QQ-P-416 - Plating, Cadmium (Electrodeposited).

PPP-B-585 - Box, Wood, Wirebound.

PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-cleated.

PPP-B-601 - Boxes, Wood, Cleated Plywood.

PPP-B-636 - Box, Shipping, Fiberboard.

PPP-B-640 - Box, Fiberboard, Corrugated Triple-wall.

PPP-B-1672 - Boxes, Shipping, Reusable with Cushioning.

MILITARY

MIL-P-116 - Preservation, Methods of.

MIL-M-3171 - Magnesium Alloy, Processes for Pretreatment & Prevention of Corrosion On.

MIL-W-5088 - Wiring, Aerospace Vehicle.

MIL-E-5400 - Electronic Equipment, Aerospace, General Specification for.

MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.

MIL-L-6085 - Lubricating Oil, Instrument, Aircraft, Low Volatility.

MIL-S-7742 - Screw Threads, Standard, Optimum Selected Series, General Specification for.

MIL-M-7793 - Meter, Time Totalizing.

MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.

MIL-C-14806 - Coating, Reflection Reducing, for Instrument Cover Glass & Lighting Wedges.

MIL-T-18303 - Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.

MIL-S-23535 - System, Integrated Flight Instrumentation.

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MIL-L-25467 - Lighting, Integral, Red, Aircraft Instrument, General Specification for.

MIL-C-26482 - Connectors, Electrical (Circular Miniature Quick Disconnect, Environmental Resisting) Receptacles & Plugs, General Specification for.

MIL-B-43666 - Box, Shipping Consolidation.

MIL-G-81704 - Glass, Aircraft Instrument, Lighting Wedge and Cover.

MIL-C-83488 - Coating, Aluminum, Ion Vapor Deposited.

MIL-L-85762 - Lighting, Aircraft, Interior, AN/AVS-6 Aviator's Night Vision Imaging System (ANVIS) Compatible.

STANDARDS

FEDERAL

FED-STD-595 - Color.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-130 - Identification Marking of U.S. Military Property.

MIL-STD-411 - Aircrew Station Signals.

MIL-STD-454 - Standard General Requirements for Electronic Equipment.

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-471 - Maintainability Demonstration.

MIL-STD-704 - Aircraft Electric Power Characteristics.

MIL-STD-781 - Reliability Tests, Exponential Distribution.

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

MIL-STD-889 - Dissimilar Metals.

DOD-STD-1866 - Soldering Process, General (Nonelectric)(Metric).

MIL-STD-2073 - DOD Material Procedures for Development & Application of Packaging Requirements.

MIL-STD-2074 - Failure Classification for Reliability Testing.

MS17869 - Indicator, Horizontal Situation, Id-1013/A.

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MS18028 - Indicator, Attitude-Director, Remote (Ifis).

MS33558 - Numeral and Letter, Aircraft Instrument Dial, Standard Form of.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

American Society of Testing and Materials (ASTM)

ASTM-D-3951 - Packaging, Commercial.

(Copies of ASTM-D-3951 should be obtained from the American Society of Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, samples shall be subjected to first article inspection (see 4.3 and 6.4).

3.2 Parts.

3.2.1 Commercial parts. Commercial parts having suitable properties shall be used, where on the date of invitations for bids, there are not suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts and cotter pins, having suitable properties, shall be used provided:

a. They can be replaced by standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts. With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suited for the purpose intended.

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3.3.1 Critical materials. Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Non-magnetic material. Non-magnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.3 Protective treatment. When materials are used in the construction of the indicator which is subjected to corrosion in environmental conditions likely to occur during normal service life, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of environmental conditions, shall not be used.

3.3.3.1 Metals. Metals shall be of the corrosion resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt environment or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

3.3.3.2 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings or hermetic sealing.

3.3.3.3 Magnesium alloy parts. Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the acquiring activity shall be used.

3.3.3.4 Aluminum alloy parts. Where practicable, aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625. Dial plates, small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.3.3.5 Iron and steel parts. Iron and steel parts which are in hermetically sealed cases shall be cadmium or aluminum plated in accordance with QQ-P-416 or MIL-C-83488. Parts which are not in sealed cases shall be chromium, nickel or aluminum plated in accordance with QQ-C-320, QQ-N-290 or MIL-C-83488, respectively. Parts in confined space in the presence of organic material shall be tin-cadmium or aluminum plated.

3.3.3.6 Fungus-proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the acquiring activity. If used in a hermetically sealed case, fungicidal treatment will not be necessary.

3.3.3.7 Fumes and vapors. Materials used in the construction of the indicator shall not produce corrosive, deleterious or toxic fumes or vapors under the conditions specified herein.

3.3.3.8 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

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3.3.4 Lubrication. The lubricating oil used in the indicator shall conform to MIL-L-6085. The amount of lubricating oil used shall be sufficient for at least 1000 hours of operation.

3.4 Design and construction. The indicator shall be designed and constructed so that the complete mechanism consisting of all components, except the pitch knob assembly and lighting components, as required, shall be removable from the case as a unit. The design shall be reasonably simple to permit service, maintenance, repair or overhaul and shall comply with the response characteristics and perform as required when subjected to the tests specified in section 4. Further, the indicator shall be constructed to have a high degree of trouble-free and adjustment-free operation with no parts working loose in service and shall withstand normal shocks, vibraton and such other conditions as are incident to service, shipping, storage and installation, without functional failure. The indicator shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.4.1 Maintainability. The mean corrective maintenance downtime (Mct) shall be 100 minutes and the maximum corrective maintenance downtime (Mmaxct) shall be not more than 300 minutes. These times include the time to unseal the indicator and hermetically seal it again after repair.

3.4.2 Reliability. The indicator shall operate with satisfactory performance, continuously or intermittently for a period of not less than 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.2.1 Reliability in mean time between failure (MTBF). The indicator shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.4.3 Longevity. The indicator shall have a total operating life of 5000 hours with minimal servicing and replacement of parts. (see 4.7 and 4.7.3).

3.4.4 Case. The case shall conform to MS17869, shall be made of non-ferrous, low density metal, uniform in texture, with a smooth surface and shall be hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service. The pitch trim knob shall be .75 inch in diameter and one inch long. The case shall be isolated from all electrical circuits within it.

3.4.4.1 Case markings. All markings on the outside of the instrument case shall be made, unless otherwise specified, with lusterless white material of Color No. 37875 of FED-STD-595. The style and proportions of the letters and numerals shall conform to MS33558. The marking material shall be of a quality sufficient to ensure durability of the markings in service use.

3.4.4.2 Hermetic sealing. The case shall provide a hermetically sealed enclosure for the entire mechanism. The case shall be so constructed that it may be opened, the mechanism removed and replaced and the case resealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any atmosphere to which the instrument may

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be subjected in normal use in military aircraft (see 4.6.18).

3.4.4.3 Filling medium. The filling medium shall be a mixture of 90 percent nitrogen and 10 percent helium. The nitrogen shall be in accordance with BB-N-411, Type I, Grade A. The filling medium shall be 98 percent pure, free of dust particles and containing not more than 0.006 mg of water vapor per liter (dew point -65°C) at the filling pressure. The case shall be filled to an absolute pressure of 1 ±0.1 atmosphere.

3.4.4.4 Cover glass. The cover glass shall be flat and free from discoloration, scratches and striae which interfere with the normal reading of the indicator and shall comply with MIL-G-81704, Type I. The cover glass shall not exceed 0.25 inch thickness. All reflecting glass surfaces shall be provided with a reflection reducing coating in accordance with MIL-C-14806.

3.4.5 Total weight. The total weight of the indicator shall be not greater than 10.0 pounds.

3.4.6 Time totalizing meter. The indicator shall contain a time totalizing meter in accordance with MIL-M-7793.

3.4.7 Cables and connectors. The indicator shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.4.8 Interconnection cabling. The indicator shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the indicator.

3.5 Interchangeability of reordered equipment. For reordered equipment, interchangeability shall exist between units and all replaceable assemblies, subassemblies and parts of a designated model of any previously manufactured equipment supplied or designated by the procuring activity.

3.6 Electromagnetic interference control. The generation of electromagnetic interference by the indicator and the vulnerability of the indicator to electromagnetic interference shall comply with the requirements of MIL-STD-461 and MIL-STD-462 (see 4.6.31).

3.7 Maintenance provisions and field testing. Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.8 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

a. Temperature	Room ambient (25°C ±5°)
b. Attitude	Normal ground

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c. Vibration	None
d. Humidity	Room ambient up to 90% relative humidity
e. Input power voltage	1. 115 ±1.0 VAC, 400 Hz 2. 26 ±1.0 VAC, 400 Hz 3. 5 ±1.0 V Lighting

3.9 Service conditions. The indicator shall operate satisfactorily under any of the environmental service conditions or reasonable combinations as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.10 Electrical requirements (see 4.6.21, 4.6.22, 4.6.30).

3.10.1 Primary input power requirements. The indicator shall meet applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with the characteristics as defined in MIL-STD-704 having limits as specified herein. The power requirement shall not exceed specified amounts.

a. AC Power (Single Phase) 115V, Category B, 55 VA
b. AC Power (Single Phase) 26V, Category B, 8 VA
c. AC Power (Single Phase) 5V, Category B, 1 VA

3.10.2 Undervoltage protection. The indicator shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within the specified limits.

3.10.3 Overload protection. Overload protection shall be provided in the indicator. All parts and circuits which are likely to carry an overload due to failures or poor adjustments shall have suitable protective devices or, preferrably, shall be protected to withstand such overload without damage to the indicator. The indicator shall resume normal operation when the overload has been eliminated.

3.10.4 Electronic parts. Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists in accordance with MIL-E-5400 shall not be used unless approved by the acquiring activity. Electronic devices and their applications shall be chosen in accordance with applicable MIL-STD-454 requirements as follows:

Requirement 30 - Semiconductor devices

3.11.5 Electrical connectors. The electrical connectors used shall conform to MIL-C-26482 and MS17869.

3.10.6 System operation. The indicator specified herein shall operate satisfactorily as part of the integrated flight instrumentation system meeting the applicable requirements of MIL-S-23535 when specified in the contract (see 6.1).

3.11 Degraded performance. Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the indicator.

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3.12 Performance. Unless otherwise specified herein or in the contract, values set forth to establish requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerences or values setting forth acceptable variations from performance under standard conditions will be specified.

3.13 Detail requirements. The indicator shall present the following information:

a. Stabilized magnetic compass heading provided from the master directional reference.

b. Command heading information provided from a remote source or desired heading set in manually.

c. Command course provided from a remote source or desired course set in manually displayed by a pointer and a numerical indication.

d. Course bar which indicates displacement of the aircraft from a manually or automatically selected track.

e. To-from indication from a selected radio facility.

f. Numerical identification of distance to a transmitting radio source or target; indication masked when not in use.

g. Bearing information presented by two independent rotating pointers, read against a compass card.

h. Power warning indication.

i. Course-bar flag alarm.

j. Bearing pointers and mode windows.

3.13.1 Miniaturization. Modern techniques of miniaturization, plotting and hermetic sealing shall be exploited to the greatest extent practicable, increasing ruggedness, reliability and service life.

3.13.2 Amplifiers. Five amplifiers provided as a part of the indicator shall be designed with redundant circuits such that the indicator will continue to operate, though with reduced performance, should partial failure of the amplifier occur. All five channels may be constructed on a single circuit board assembly.

3.13.3 Integral lighting. The indicator shall have red integral lighting in accordance with the requirements of MIL-L-25467. The mode light signals of the indicator shall be in accordance with the warning lights requirements of MIL-STD-411. The case shall not be used as grounding for the lighting circuit (see 4.6.2 and 4.6.19).

3.13.3.1 ANVIS compatible lighting. When required as specified in the contract or purchase order, aviator night vision imaging system (ANVIS) conpatible lighting shall be in accordance with MIL-L-85762 (see 4.6.19.1 and 6.2.1).

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3.13.4 Visibility of dial. The pointers, numerals, indexes and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpindicular to the dial and whose small diameter is the aperture of the case.

3.13.5 Dial marking. The scale and other markings on the dial shall be as shown on Figures 1 and 2, including the dimensions of the markings. The style and proportions of the letters and numerals placed on the dial shall conform to MS33558. All markings shall be made with lusterless white material of Color No. 37875 of FED-STD-595.The background of the dial shall be finished with lusterless black material of Color No. 37038. The dial marking material shall be of a qualitiy sufficient to ensure durability of the markings in service use.

3.13.6 Numerals. Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be placed so that the center of the mass of the numeral is on the radial line joining the appropriate graduation and the center of the dial. Any confusion resulting in doubt as to which graduation the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64th inch.

3.13.7 Presentation. The presentation shall be as shown on Figure 2 and shall consist of a rotating-card type azimuth ring, a lubber line, a heading marker, a course arrow, a to-from arrow, a course bar, two bearing pointers, a selected course window, a distance window, a power-off warning flag, a course bar flag alarm and a miniature airplane symbol behind the cover glass of the instrument to present a pictorial display of the navigation situation.

3.13.8 Lubber line and airplane. The lubber line and the airplane shall be as shown on Figures 1 and 2. The airplane may be painted on the inside of the lighting wedge to comply with this requirement.

3.13.9 Aircraft heading The aircraft heading shall be displayed on the azimuth ring and shall read under the fixed lubber line located at the top of the indicator face as shown on Figures 1 and 2.

3.13.10 Azimuth ring signal input. The input synchro control transformer shall be a high impedance type EP AY500-5, or electrical equivalent, which shall receive an input from a synchro device similar to the standard synchro test transmitter described in section 4.

3.13.11 Azimuth ring servo system. A servo follow-up system consisting of a control transformer, motor and necessary feedback and amplifier shall drive the azimuth card and rotor of the input control transformer to a null. It shall be capable of receiving a change of heading at any rate between 0 and 60°/sec. without hunting. There shall be no appreciable hunt or oscillation in the servo system while the input compass signal is changing or stationary. The servo system shall position the heading indication, under the lubber line, to within $\pm$.5° of the position of a standard synchro test transmitter (see 4.6.30). The design of the azimuth card gearing shall result in operation as would be achieved in the system shown on Figure 3. The electrical pin connections of the azimuth card servo system shall conform to that shown on Figure 3 (see 4.6.4).

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3.13.12 Azimuth card control transformer zero. The azimuth card control transformer shall be zeroed with a positive rotation reference at an index reference of 270°. With the 'X', 'Y' and 'Z' leads of the azimuth card control transformer as shown on Figure 3 and the excitation voltage of the standard synchro test transmitter 'C' to 'H' in phase with the power input pin 'A' to pin 'B':

a. The azimuth ring shall position at 0° (north) when the rotor of the standard synchro test transmitter dial is at 0°.

b. If the dial on the rotor of the standard synchro test transmitter is continuously rotated such that its heading indication increases, the azimuth ring heading shall increase (counterclockwise rotation) accordingly, and voltage 'C' to 'H' of the control transformer shall be in phase with the excitation voltage 'C' to 'H'.

3.13.13 Command heading. A means of selecting a desired heading both automatically and manually shall be provided. A heading marker shown on Figures 1 and 2, which can rotate around the azimuth ring, shall be driven by a servo system when excited by a remote device similar to the standard synchro transmitter specified in section 4 and manually by a knob at the lower left corner of the indicator. There shall be no visible interaction between the knob and the servo motor such that when the remote synchro device is driving the servo system, turning of the knob should have no lasting effect on the heading marker position. The heading marker shall rotate in synchronicity with the azimuth ring after it is set. The heading knob shall be connected with the heading marker in such a manner that clockwise rotation of the heading marker will produce clockwise rotation of the course arrow, and vice-versa. The turning ratio between the knob and the heading marker shall be between 8 and 10 to 1. The azimuth scale reading indicated under the heading marker shall be the desired heading. The heading set knob shall be as shown on Figure 1. The backlash between the command heading marker and the azimuth card shall be not greater than .25°.

3.13.14 Heading marker servo system. A servo system similar to that described in 3.14.11 for driving the azimuth ring shall be employed for positioning the heading marker in response to a remote input signal. The servo shall be capable of receiving a change of command heading at any rate between 0 to 60°/sec without hunting. There shall be no appreciable hunt or oscillation in the servo system and it shall position the heading marker around the azimuth ring to within one degree of the position of the test synchro. (The design of the differential gearing shall result in operation of the heading marker with respect to the azimuth ring, remote synchro command input and knob command input as that shown on Figure 3). The heading marker servo system shall conform to the electrical connector and pin wiring shown on Figure 3.

3.13.15 Heading marker signal input. The heading marker servo system shall receive its input signal from a transmitting synchro device similar to the standard synchro test transmitter described in section 4. The heading synchro device shall be capable of operation as a control transformer and a transmitter as shown in Figure 3. The 'X', 'Y' and 'Z' load shall be equivalent to an EP AY-500-3.

3.13.16 Heading datum control transformer. The heading datum control transformer shall have the general impedance characteristics of an EP AY-500-5

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or equal and operate as shown on Figure 3.

3.13.17 Heading command and heading datum control transformer zero. Both the heading command and heading datum control transformers shall be zeroed with a positive rotation referenced at an index reference of 270°. Both synchros shall be zeroed in the following manner:

a. With the standard synchro test transformer positioned to zero indication on its dial and locked, and connected to the heading command and heading datum control transformer as per 'X', 'Y' and 'Z' notation indicated in Figure 3, the standard synchro test transformer shall be excited with 26 V, 400 Hz.

b. The heading marker shall be positioned to 0° (north) on the azimuth card and the voltage 'C' to 'H' of the heading command control transformer rotor and heading datum control transformer rotor as shown on Figure 3 shall be null.

c. If the heading marker is rotated clockwise around the azimuth ring, the voltage 'C' to 'H' of the heading command control transformer rotor and the heading datum control transformer rotor shall be in the opposite phase sense as the voltage 'C' to 'H' of the standard synchro test transmitter from zero (north) to 180°, null at 180° (south) and shall be in the same phase sense as the voltage 'C' to 'H' of the standard synchro test transmitter from 180° (south) to 0° (north).

d. If the heading marker is set on 0° (north) on the azimuth dial and if the rotor of the standard synchro test transmitter is rotated to indicate an increasing heading (dial counterclockwise) the voltage 'C' to 'H' of the heading command control transformer and the heading datum transformer shall be null at 0°, in the same phase sense as the excitation voltage 'C' to 'H' from 0° to 180°, null at 180° and opposite to the phase sense of 'C' to 'H' from 180 to 0°.

e. If the transmitter winding of the heading command control transformer is excited with 26 V AC such that the voltage 'C' to 'H' is in phase with the power excitation ground to phase and the heading marker is set on 'North', then the voltage 'X' to 'Z' and 'Y' to 'Z' shall be in phase with 'C' to 'H' and voltage 'X' to 'Y' null. If the heading marker is rotated clockwise to 'East', the voltage 'X' to 'Z' shall increase before decreasing.

3.13.18 Heading marker servo zero. The heading marker command control transformer shall be connected to its servo system. The standard synchro test transmitter shall be connected to the heading control command transformer as per the 'X', 'Y' and 'Z' notation on Figure 3. The excitation voltage of the standard synchro test transmitter 'C' to 'H' shall be in phase with the servo power excitation voltage, pin A to pin C. The following conditions shall be satisfied:

a. The heading marker shall position to 0° (north) on the azimuth ring when the dial of the standard synchro test transmitter is at 0°.

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b. The heading marker shall rotate in an increasing heading indication on the azimuth ring (clockwise) when the dial of the standard synchro test transmitter is rotated in an increasing heading indication (dial counterclockwise).

c. If the dial of the standard synchro test transmitter is rotated in the increasing heading indication, the voltage 'C' to 'H' of theheading command control transformer shall be in the same phase sense with the voltage 'C' to 'H' of the standard synchro test transmitter.

3.13.19 Command course. A means of selecting a desired course both automatically and manually shall be provided. A course marker shown on Figure 1 which can rotate around the inside edge of the azimuth ring, shall be driven by a servo system when excited by a remote device similar to the standard synchro transmitter specified in section 4 and manually by a knob at the lower right corner of the indicator. Turning of the course knob shall have no lasting effect on the course arrow when the servo is being driven by a remote device. The course marker shall rotate in synchronicity with the azimuth ring after it is set. The scale reading on the azimuth ring at the tip of the course arrow shall be the selected course or track. The course knob shall be connected with the course arrow in such a manner that clockwise rotation of the knob produces a clockwise rotation of the course arrow, and vice-versa. The turning ratio between the knob and the course marker shall be between 8 and 10 to 1. The course set knob shall be as shown on Figure 1. A digital-type counter shall present the selected course in a window on the right side at the top of the instrument as shown on Figure 1.

3.13.20 Course marker servo system. A servo system similar to that described in 3.14.14 for driving the azimuth ring shall be employed for positioning the course marker in response to a remote input signal. The servo shall be capable of receiving a change of command course at any rate between 0 and 60°/sec without hunting. There shall be no appreciable hunt or oscillation in the servo system during steady state and dynamic operation and it shall position the course marker around the inside edge of the azimuth ring to within ±1° of the position of the test synchro. The design of the differential gearing shall result in operation of the course marker with respect to the azimuth ring, remote synchro command input as that achieved in Figure 3. The course marker servo shall conform to the electrical connector and pin wiring shown on Figure 3.

3.13.21 Course marker signal input. The course marker servo system shall receive its input signal from a transmitting synchro device similar to the synchro test transmitter described in section 4. The course synchro device shall be capable of operating as a control transformer and a transmitter as shown in Figure 3. The 'X', 'Y' and 'Z' loads shall be equivalent to an EP AY-500-3 or one exhibiting higher impedance characteristics.

3.13.22 Course datum control transformer. The course datum control transformer shall have the general impedance characteristics of an EP AY-500-5 synchro or equal as shown on Figure 3.

3.13.23 Course command and course datum control transformer zero. Both the course command and course datum shall be zeroed as a standard synchro test transformer at an index reference of 270°. Both synchros can be zeroed in the identical manner as the heading command transformer and heading datum

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control transformer given in 3.14.12 with course marker substituted for heading marker and all reference made to the course servo system. The course servo system should not be excited with power.

3.13.24 Course marker servo zero. The course command control transformer shall be connected to the standard synchro test transmitter as per the 'X', 'Y' and 'Z' notation shown on Figure 3. The excitation voltage of the standard synchro test transmitter 'C' to 'H' shall be in phase with the servo power excitation voltage, pin 'A' to pin 'D' on the servo amplifier. The following conditions shall be satisfied:

a. The course marker shall position to 0° north on the azimuth ring when the dial of the standard synchro transmitter is at 0°.

b. The course marker shall rotate in an increasing heading indication in the azimuth ring (clockwise) when the dial of the standard synchro test transmitter is rotated in an increasing heading indication (dial counterclockwise).

c. When the dial of the standard synchro test transmitter is rotated in an increasing heading indication (dial counterclockwise) the voltage 'C' to 'H' of the course command transformer shall be in phase with the voltage from 'C' to 'H' of the standard synchro test transmitter.

d. If the transmitter winding of the course command control transformer is excited with 26 V AC such that voltage 'C' to 'H' is in phase with power excitation ground to phase and the course pointer is set on 'North' then the voltages 'X' to 'Z' and 'Y' to 'Z' shall be in phase with 'C' to 'H' and voltage 'X' to 'Y' null. If the course pointer is rotated clockwise to 'East' the voltage 'X' to 'Z' shall increase before decreasing.

3.13.25 Course resolver. A resolver of any size with electrical characteristics of a type EP AY-211-5-B shall be connected to the course gear train to function as shown on Figure 3. The resolver shall be electrically connected as shown on Figure 3 and zeroed at a scale reference of 0° and an index reference of 300°.

a. Rotor winding number 1 (RW-1) is defined as the leading wind which mechanically leads rotor winding number 2 (RW-2) by 90° in the direction of higher dial numbers with respect to the azimuth card. At the index reference point (300°) when properly zeroed stator winding 1 (SW-1) and RW-1 are directly coupled and stator winding 2 (SW-2) and RW-2 are directly coupled.

b. The common pin of the rotor windings is designated as pin 'C'. The two leads of a stator winding are arbitrarily labeled 'D' and 'E' and this stator winding is called SW-1.

c. The course marker is rotated to 300° against the azimuth card. Choose a rotor lead and temporarily label it 'A'. Excite this rotor winding ('C' being the common ground) with 10V, 400 Hz AC power. Adjust resolver rotor shaft (with scale locked at 300°) until voltage 'D' to 'E' of SW-1 is maximum and in phase with voltage 'A' to 'C' respectively. Lock resolver rotor to scale and rotor scale

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90° a positive direction to 30° against the azimuth card. Remove voltage 'A' to 'C' of RW-1 and apply it to 'B' to 'C' of RW-2. Voltage as read across 'D' to 'E' of SW-1 should be a maximum and in phase with applied voltage 'B' to 'C' of RW-2. If voltage isout of phase reverse the labeling 'A' to 'B' of the rotor leads and repeat the above zeroing procedure.

d. Set the scale reading to 300° against the azimuth card and excite leads 'B' and 'C' of RW-2. Assign letters 'F' and 'G' to the stator winding SW-2 such that the voltage measure from 'F' to 'G' is in phase with voltage 'B' to 'C' of RW-2.

e. Fine zeroing can be accomplished by exciting leads 'B' to 'C' of RW-2 and nulling the voltage 'D' to 'E' of SW-1 by rotating resolver rotor shaft when scale reading is 300° against the azimuth card.

f. The course selector resolver shall have proper characteristics or additional provisions as required to permit use with VOR systems requiring a 30 Hz omni range zero. Proper omni range zero operation at 30 Hz shall be obtained when the resolver is electrically zeroed with 400 Hz excitation at 300°. The omni range zero rotor leads shall be designated as leads 'H' and 'C'. The leads designated above as 'A' to 'G' shall be connected as specified in Figure 3. If a separate choke is added, it shall be given the lead notation of 'H' and shall be connected to pin 'A' as shown on Figure 3.

NOTE: For 400 Hz operation, pins a, b, c, d, e, f and g are utilized to obtain the sine and cosine relationship. For 30 Hz omni range zero operation with a separate choke, pin 'A' is activated and pins a and b are not.

3.13.26 Meter movement bearings and pivots. Meter movements shall turn on hardened ground and highly polished pivot bearings, highly polished first quality sapphire bearings, or other materials that have been approved by the procuring activity (see 4.6.5).

3.13.27 Balancing. The meter movements shall be balanced by an approved means (threaded nut or splice nuts, or wire helix moving along a crown arm and a counterweight arm) with easily accessible means for rebalancing. The use of solder, shellac or similar means of holding balance weights will not be accessible (see 4.6.10).

3.13.28 Power failure indicator. Power failure warning indication shall be incorporated on the indicator as shown on Figures 1 and 2. The word 'OFF' shall be black on white background. A lead shall be brought out through the electrical connector as shown on Figure 3. When the ground connection is removed or power is removed the warning shall appear. When power is applied but internal failure of the b+ amplifier occurs, as shown on Figure 3, the warning shall appear (see 4.6.16).

3.13.29 Course bar. An indication of the displacement of the aircraft to be right or left of the selected course shall be provided by means of a course bar and dots as shown on Figures 1 and 2.

3.13.30 Course bar mechanism. Direct current deflection signals cause the course bar to move across the face of the indicator. The total movement of the

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course bar shall be ±11/16 inch measured from the center of the indicator. A movement of .625 inch from center shall require 150 microamperes ±7.5%. A direct current polarity of (+) and (-) indicated as pins of Figure 3 with connector J-1 and pins u(+) and t(-) shall cause a deflection to the right when course arrow is on top. The course bar sensitivity shall be 75 microamperes/dot and range of displacement shall be ±2 dots. The course bar mechanism shall have a resistance of 1000 ohms ±3% and a damping ratio not less than 1.0. The frequency response of the course bar mechanism shall be such that amplitude ratio of output to input shall not decrease by more than 3 decibels at 0.2 Hz (see 4.6.9 and 4.6.11).

3.13.31 Course bar flag alarm. This flag alarm shall be mounted as shown on Figures 1 and 2, and shall rotate with the central portion of the display. The flag alarm shall be operated by a suppressed zero type of mechanism which shall hold the flag against a stop in the position shown on Figures 1 and 2 in the absence of current or when the current application is below a predetermined value as specified herein. The flag shall be designed to move out of view when the input signal polarity as shown in Figure 3 with connector J-2 and pins P(+) and R(-). The flag display shall be white when in view and black when properly activated. The flag shall leave its visible position stop with a current application of 180 microamperes or more. The flag shall disappear completely from view with a current application of 245 microamperes. Response of the flag alarm mechanism shall be linear with respect to current. The application of 500 microamperes as a step input shall not damage the meter movement. The resistance across the flag input shall be 1000 ohms ±3% (see 4.6.13).

3.13.32 To-from arrow. A to-from arrow shall be incorporated as shown on Figure 1. The to-from arrow shall operate from a meter movement having a resistance of 250 ohms maximum and 150 ohms minimum. The sensing will be such that with no signal applied the arrow will not be visible. A signal with polarity as shown on Figure 3 with connector J-1 and pins u(+) and v(-) shall cause the to-from arrow to appear on the same side as the course arrow. A signal with opposite polarity shall cause the to-from arrow to appear on the tail side of the course arrow. A signal not greater than 225 microamperes shall cause the to-from arrow to reach full displacement (se 4.6.14).

NOTE: The central portion of the display including the course arrow, course bar, to-from arrow and course deviation dots shall rotate as a unit turning with the azimuth ring as the airplane's heading changes, in addition to the individual motions listed.

3.13.33 Bearing markers #1 and #2. Bearing marker servo systems shall be incorporated for positioning the pointers #1 and #2 shown on Figures 1 and 2 from remote sources similar to a standard synchro test transmitter. The servo systems shall be similar to that driving the azimuth ring. The servos shall be capable of receiving a change in bearing between 0 to 60°/sec without hunting. There shall be no appreciable hunting or oscillation in the servo systems and they shall position the bearing pointers around the azimuth ring to within ±1° of the position of the standard synchro test transmitter. The servo systems shall be connected as shown on Figure 3.

3.13.34 Bearing pointer signal input. Each bearing pointer servo system shall receive its input signal from a transmitting synchro device similar to the standard synchro test transmitter described in section 4. The bearing pointer control transformer shall be a type EP AY-500-5 synchro or equal.

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3.13.35 Bearing pointer control transformer zero. Each bearing pointer control transformer shall be zeroed at a scale reference of 180° (under the lubber line at the bottom of the display) and index reference of 90°. With the standard synchro test transmitter connected as per the 'X', 'Y' and 'Z' notation on Figure 3 and excited with a voltage from 'C' to 'H' in phase with the servo excitation pins 'A' to 'B' the transmitter dial shall be positioned to zero.

a. The bearing pointer shall position under the lubber line at the bottom of the display.

b. Rotation of the card on the standard synchro test transmitter to an increasing heading indication shall cause the bearing pointer to rotate clockwise.

c. If the rotor of the standard synchro test transmitter is rotated to an increasing heading indication the voltage 'C' to 'H' of the bearing control transformer shall be in the same phase as the voltage 'C' to 'H' of the standard synchro test transmitter.

3.13.36 Distance display. A digital-type distance display shall be provided at the upper left of the indicator as shown on Figures 1 and 2. The range of the display shall be 000 miles to 999 miles. The hundreds, tens and units numerals shall be individually driven by receiver synchros with a 1:1 ratio. The units display shall have one mile and .5 mile index lines. The numerals shall be centered each 36° of synchro shaft rotation. The receiver synchros shall be Clifton CRS-S-A-1 or equal. The thousand digit shall be driven by a shutter showing black normally unexcited and '1' when excited by 27 V DC (see 4.6.15).

3.13.37 Distance synchro electrical zero. With a standard synchro test transmitter connected as per the 'X', 'Y', 'Z' common with 'C' notation and 'C' to 'B' of both the receiver synchros and transmitter synchro excited, the three synchros shall be zeroed as follows:

a. With standard synchro test transmitter rotor set on 0°, the '0' shall be in position on the distance dial of each receiver synchro.

b. Rotation of the standard synchro test transmitter to 36° shall result in '1' being positioned on the distance dial.

c. Increasing numerical dial readings on the receiver synchros shall be indicated for each 36° of the transmitter synchro rotation.

3.13.38 Distance shutter mechanism. A black masked shutter shall be provided for obscuring the distance display. The shutter mechanism when excited by 27 V DC voltage excitation shall cause the shutter to disappear from view. The units digit shall be only partially masked by the shutter (see 4.6.17).

3.14 Examination of product. The indicator, when tested as specified in 4.6.1, shall operate without failure and show no structural degradation.

3.15 Synchro and resolver zero. The indicator, when tested as specified in 4.6.3, shall operate without failure.

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3.16 Pointers and dial sensitivity. The indicator, when tested as specified in 4.6.6, shall operate without failure.

3.17 Scale error. The indicator, when tested as specified in 4.6.7, shall operate without failure.

3.18 Servo follow-up and rate and damping. The indicator, when tested as specified in 4.6.8, shall operate without failure and should move smoothly throughout its range.

3.19 Pointer freedom. The indicator, when tested as specified in 4.6.12, shall operate without failure and show no structural degradation.

3.20 Manufacturer run-in test. The indicator, when tested as specified in 4.6.20, shall operate without failure and show no structural degradation.

3.21 Low temperature operation. The indicator, when tested as specified in 4.6.23, shall operate without failure.

3.22 High temperature operation. The indicator, when tested as specified in 4.6.24 and 4.6.32, shall operate without failure.

3.23 Fogging. The indicator, when tested as specified in 4.6.25, shall operate without failure and show no structural degradation.

3.24 Magnetic effect. The indicator, when tested as specified in 4.6.26, shall operate without failure.

3.25 Acceleration. The indicator, when tested as specified in 4.6.27 and 4.6.35, shall operate without failure.

3.26 Dielectric strength. The indicator, when tested as specified in 4.6.28, shall operate without failure.

3.27 Vibration. The indicator, when tested as specified in 4.6.29 and 4.6.36, shall operate without failure.

3.28 Low pressure. The indicator, when tested as specified in 4.6.33, shall operate without failure.

3.29 Temperature shock. The indicator, when tested as specified in 4.6.34, shall operate without failure.

3.30 Impact shock. The indicator, when tested as specified in 4.6.37, shall operate without failure.

3.31 Humidity. The indicator, when tested as specified in 4.6.38, shall operate without failure.

3.32 Salt fog. The indicator, when tested as specified in 4.6.39, shall operate without failure.

3.33 Fungus resistance. The indicator, when tested as specified in 4.7.8, shall operate without failure.

3.34 Identification of product.

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3.34.1 Nameplate. A nameplate shall be securely attached to the exterior of the case and shall be durably marked to withstand usage encountered in service. The nameplate shall be legibly marked in accordance with MIL-STD-130.

3.34.2 Manufacturer's part number. The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.35 Workmanship. The indicator, including all parts and accessories, shall be so constructed and finished that it shall be free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, plating, painting, riveting, machine screw assemblies and freedom of parts from burrs and sharp edges (see 4.7.9).

3.35.1 Dimensions. Each indicator shall be checked dimensionally to determine conformance to the dimensions in MS18028. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the instrument, they shall be held or limited accordingly.

3.35.2 Soldering. Soldering shall be performed in accordance with DOD-STD-1866.

3.35.3 Screw threads. Screw threads 0.0060 inch in diameter or larger, shall be in accordance with MIL-S-7742.

3.35.4 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Backlash shall be held to a minimum consistent with the application of the gear assembly.

3.35.5 Cleaning. The indicator shall be thoroughly cleaned and free from loose, splattered or excess solder, resin flash that may crumble, metal chips and other foreign matter after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring all products or supplies submitted to the Government for acceptance comply with all

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requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test procedures. The procedures used for conducting all required tests shall be prepared by the contractor. The right is reserved by the Government to modify the test or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved tests are available from previous contracts, such procedures will be provided and may be used. However, the right is reserved by the acquiring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall consist of all of the tests of 4.6 of this specification (see 6.4).

4.3.1 First article test samples. First article test samples shall consist of three indicators manufactured in accordance with this specification. The indicators submitted for first article testing shall not have been previously tested except for the individual tests. The samples shall be forwarded to the testing laboratory designated by the acquisition activity.

4.3.1.1 First article test sample identification. The test samples shall be plainly identified by securely attached durable tags marked with the information listed below:

Sample for First Article Inspection
Horizontal Situation Indicator ID-1013( )/A
Submitted by (manufacturer's name, date) for first article inspection in accordance with MIL-I-23366B under contract number
MS -(add proper dash number)
Manufacturer's part number

4.4 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing all tests except sampling plan B tests which will be conducted at a laboratory designated by the acquiring activity. Quality conformance tests, except for sampling plan B, shall be under the supervision of the Government quality control representative. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following:

a. Individual tests (see 4.4.1)

b. Sampling plan A tests (see 4.4.2.1.1)

c. Sampling plan B tests (see 4.4.2.2.2)

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d. Reliability tests (see 4.4.4)

4.4.1 Individual tests. Each indicator submitted for acceptance shall be subjected to the individual tests listed below. These tests shall determine compliance with the requirements of design, construction, workmanship and operational accuracy. When the indicator fails any one of the tests at any point, the indicator shall be rejected and no further test shall be performed on the item. As a minimum, each indicator shall have passed the following tests:

a. Examination of product (see 4.6.1)

b. Synchro and resolver zero (see 4.6.3)

c. Servo operation (see 4.6.4)

d. Servo follow-up rate and damping (see 4.6.8)

e. Course bar mechanism (see 4.6.9)

f. To-from arrow (see 4.6.14)

g. Distance display (see 4.6.15)

h. Power failure warning (see 4.6.16)

i. Shutter (see 4.6.17)

j. Case leakage (see 4.6.18)

k. Lighting (see 4.6.19)

l. Manufacturing run-in test (see 4.6.20)

4.4.2 Sampling plan tests. The sampling plan tests shall consist of sampling plan A and sampling plan B tests. Indicators selected for sampling tests shall first have passed the individual tests. Indicators which have been subjected to the sampling plan A tests shall not be delivered on contract until they have been refurbished (see 6.3.3) and resubmitted for all the individual tests.

4.4.2.1 Sampling plan A sample selection. Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity of indictors offered for acceptance	Quantity selected for testing
First 15	1*
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

*When sampling plan B is invoked, the quantity is 0.
When sampling plan B is omitted, the quantity is 1.

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NOTE: When a defective indicator is detected, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs shift to one sample of the next fifteen and proceed as indicated.

4.4.2.1.1 Sampling plan A tests. Each indicator selected for sampling plan A testing shall be subjected to the tests listed below:

a. Individual tests (see 4.4.1)

b. Power consumption (see 4.6.21)

c. Voltage and frequency variation (see 4.6.22)

d. Low temperature operation (see 4.6.23)

e. High temperature operation (see 4.6.24)

f. Fogging (see 4.6.25)

g. Magnetic effect (see 4.6.26)

h. Acceleration sampling plan A (see 4.6.27)

i. Dielectric strength (see 4.6.28)

j. Vibration functional (see 4.6.29)

k. Acceleration sampling plan B (see 4.6.35)

l. Low voltage (see 4.6.30)

4.4.2.2 Sampling plan B instructions. Two indicators shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These indicators shall be forwarded to a laboratory designated by the acquiring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information:

INDICATOR, HORIZONTAL SITUATION ID-1013( )/A
Submitted by (manufacturer's name, date)
for quality conformance sampling plan B
testing, in accordance with contract
Manufacturer's part number

4.4.2.2.1 Sampling plan B approval. Approval of sampling plan B indicators shall be by the acquiring activity upon satisfactory completion of the designated tests. Any material, design or performance defect made evident during these tests shall be corrected by the contractor to the satisfaction of the acquiring activity. Failure of the sample units to pass any of the tests shall be cause for delivery of indicators under the contract to cease until proper corrective action is approved and accomplished.

4.4.2.2.2 Sampling plan B tests. Each indicator selected for sampling plan B testing shall be subjected to the tests listed below:

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a. Sampling plan A tests (see 4.4.2.1.1)

b. Electromagnetic interference (see 4.6.31)

c. High temperature exposure (see 4.6.32)

d. High altitude, low temperature (see 4.6.33)

e. Temperature shock (see 4.6.34)

f. Acceleration sampling plan B (see 4.6.35)

g. Vibration endurance (see 4.6.36)

h. Impact shock (see 4.6.37)

i. Humidity (see 4.6.38)

j. Salt fog (see 4.6.39)

k. Structural test (see 4.7.9)

4.4.3 First article tests. Three indicators shall be produced by the contractor representative of the indicators produced under contract and subject to the following first article tests:

a. Sampling plan B tests (see 4.4.2.2.2)

b. Life test (see 4.7)

c. Fungus resistance (see 4.7.8)

4.4.4 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the qualification (first article) phase and reliability assurance production acceptance test phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074.

4.4.4.1 First article phase. Prior to the acceptance of indicators under the contract or order, a minimum of three and a maximum of six indicators shall be tested as outlined in MIL-STD-781 under the section entitled "First Article Phase of Production Reliability Tests." For the first article phase, test level F shall be used. Accept-reject criteria for test plan III shall be used.

4.4.4.2 Reliability assurance production acceptance test phase. Reliability assurance production acceptance test phase testing shall be conducted through a burn-in. Each indicator produced shall pass a minimum of 100 hours burn-in test. Any failure occurring during the last 25 hours which requires component changes or major recalibration, shall be subjected to an additional 25 hours burn-in after completion of repairs. (The 100 hours shall be completed plus an additional 25 hours, a total of 125 hours.) Indicators accumulating 200 hours or more shall be overhauled (bearings replaced and other wear parts replaced as necessary) and resubmitted to a minimum 100 hours burn-in test. The indicators shall be subjected to and shall meet the requirements of the individual tests after completion of the burn-in test. During the burn-in

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period, the equipment shall be subjected to the following environmental conditions:

a. Temperature. -54°C to +55°C (-65°F to +131°F)

b. Temperature cycling. Temperature cycling shall be the time to stabilize at low temperature followed by time to stabilize at high temperature, plus two hours at high temperature.

c. Vibration. 2.2 g +10-0% peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of operating time.

d. Equipment on-off cycling. Equipment off during cooling cycle and on during heating cycle. Only the on time shall be used for the minimum 100 hours.

e. Input voltage. Nominal specified voltage +5-2%.

4.4.4.3 Performance characteristics to be measured. During the final cycle and while at -40°C and again while at 54°C each indicator shall meet the requirements of synchro resolver zero (4.6.3), servo operation (4.6.4), servo follow-up (4.6.8) course bar mechanism (4.6.9), to-from arrow (4.6.14), distance display (4.6.15), power failure warning (4.6.16), shutter (4.6.17) and lighting (4.6.19). The operating test hours and failures shall be accumulated and the results compared with the accept/reject line of test plan XVIII of MIL-STD-781. (Extend the line as necessary to accomodate the test data.) These totals shall reflect the data accumulated for a contract from the beginning of the production reliability assurance test to the present. At any time that the current totals of test hours and test failures plotted on test plan XVIII curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop acceptance of equipment whenever a reject situation exists. No indicator shall be shipped that has not passed the production reliability acceptance test and is in "good as new" condition (see 6.3.2).

4.4.4.4 Thermal survey. A thermal survey shall be conducted of the indicator to be tested, utilizing the temperature and duty cycling for the required test level prior to initiation of reliability testing. The thermal survey will identify temperature relationships between identified components of greatest thermal inertia and the chamber air temperature. These relationships shall be used for determining equipment thermal stabilization during the test. The lowest test level temperature stabilization takes place when the rate of change of the point of maximum thermal inertia at the upper temperature limit is less than 2°C/hr. Temperatures of the heating-cooling air shall be recorded continuously during both the survey and testing. The equipment thermal survey needs to be made only once for each identical equipment type representative of the manufacturer's normal production under the current production.

4.4.4.5 Test chambers. Test chambers shall be capable of maintaining the ambient and forced air temperatures at the specified test level temperatures to within ±2°C during the test. The rate of temperature change of the thermal medium in both heating and cooling cycles shall average not less than 5°C/min. Chamber and forced air temperatures shall be monitored

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continuously. A method to interrupt the automatic cycling to allow for a dwell at the minimum and maximum air temperatures shall be provided.

4.4.4.6 Duty cycle. The test equipment and fixtures shall include provision for simultaneous, vibration and temperature cycling. During testing, input signals shall be applied to the indicator under test so as to cycle the bearing pointers, compass card and distance presentation at a rate of 20°/sec. The direction of rotation shall be reversed every two minutes. When the compass card card is rotating in the counterclockwise direction the shutter shall be out of view. The indicator lighting shall be energized for 57 minutes of each hour and off for three minutes. When the indicator is off, the flag alarm shall appear.

4.4.4.7 Failure criteria. The failure criteria for the qualification phase and the all equipment production reliability acceptance test phase shall be whenever performance characteristics fail to meet the acceptance requirements (see 4.4.4.3) or any other obvious defect. In case of failure, the failure shall be recorded and the test interrupted while repairs are made. Upon completion of each 250 hours of temperature cycling, each indicator shall meet the requirements of the individual tests. No recalibration shall be performed on the indicators during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report, in accordance with MIL-STD-781, which shall contain test procedures, details of any failures and corrective actions taken. Should failure occur during the reliability test, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or a design defect.

c. Include a test under the individual inspection (4.4.1) to check all indicators for this requirement until assurance is obtained that the defect has been corrected.

4.4.4.8 Reliability assurance production acceptance phase. Whenever performance characteristics fall below the acceptance requirements of 4.4.4.3, at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure unless one part caused the other parts to fail.

4.4.4.9 Preventive maintenance. No preventive maintenance may be accomplished on the samples while they are under test.

4.4.4.10. Operational stability. Adjustments to the indicator under test shall not be performed after the reliability test has been initiated in order to meet the specification performance requirements.

4.4.4.11 Disposition of samples upon completion of tests. Any sample tested may be delivered on contract provided it meets all the following requirements:

a. It is representative of production indicators currently being accepted.

b. It is in "good as new" condition or has been referbished (see 3.2, 6.3.2 and 6.3.3).

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c. It is otherwise satisfactory.

4.5 Inspection conditions.

4.5.1 Standard conditions. Unless otherwise specified herein or in the contract, all tests required by this specification shall be made under the following conditions:

a. Temperature - Room ambient 25°C ±5°

b. Pressure - Normal atmospheric (approximately 29.92 inches Hg)

c. Vibration - None

d. Humidity - Room ambient up to 90% relative humidity

e. Input voltage - 115 ±1.0 V AC

4.5.2 Test readings. Unless otherwise specified herein or in the contract, the indicator shall be vibrated before a test reading is taken. An electrical vibrator set at 30, 60 or 120 Hz with amplitude of 0.002 inch when attached to the test item, shall be used.

4.5.3 Attitude. Unless otherwise specified herein or in the contract, the indicator shall be tested in its normal operating position with the dial vertical.

4.5.4 Lighting. Unless otherwise specified herein or in the contract, the indicator lighting system shall be energized and shall operate satisfactorily during and after all tests except humidity, fungus resistance and salt fog.

4.5.5 Standard test transmitter. The standard test transmitter shall be an Eclipse-Pioneer Type Ay-201-1, or equal, with minimum and known errors. These errors shall be subtracted from those of the indicator under test. The indicator shall have an accurate dial equipped with a ± degree vernier and readable to 0.1°. The dial shall be graduated through 360° and shall be capable of continuous rotation. The transmitter rotors shall be excited from the same power supply voltage which operates the indicator. Excitation shall be applied so that the voltage from pin 'A' to pin 'B' of the indicator with lead 'C' and A common. The transmitter shall be calibrated and the rotor and stator leads identified as follows:

a. Designate the rotor leads as 'H' and 'C', 'C' to be grounded.

b. Designate one stator lead 'Z' and connect it to 'C'.

c. Apply 26 V, 400 Hz across 'H' and 'C' and measure the voltage across the remaining stator leads and the voltage across 'H' and one of the stator leads.

d. Turn the rotor until the voltage across 'H' and one stator lead is approximately maximum (about 35 V) and the voltage across the two stator leads is exactly minimum. Lock the dial to the rotor with the dial exactly on zero.

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e. Turn the dial through increasing numbers and measure the voltages across the two undesignated stator leads and 'Z'. The voltage across 'XZ' will increase before decreasing and the voltage across 'YZ' will decrease before increasing.

4.5.6 Proper connection. When the indicator is referred to as being properly connected, it shall be understood that it is connected to a test fixture which is providing a lighting circuit voltage variable from 5 V (AC or DC) to approximately 0 V, synchro signals in accordance with 3.4.5.1 through 3.4.5.7 variable current DC signals 1000 ohm source and with reversible polarity to the meter movements. AC power as required, and the pin 'J' is jumped to pin 'A' through a closed switch.

4.6 Inspection methods.

4.6.1 Examination of product. Each indicator shall be examined to determine conformance with specification requirements for dimensions, weight, color, markings, nameplate and workmanship (see 3.14).

4.6.1.1 Accuracy test during environmental tests. The accuracy test required in 4.6.11 through 4.6.17 shall be performed as follows:

a. Compass card at 0, 90, 180 and 270° in both CW and CCW directions.

b. Pointers #1 and #2 at 180, 270, 0 and 90° in both CW and CCW directions.

c. Distance presentation at 1-3-7 numbers for units, tens and hundreds.

4.6.2 Illumination of display. The indicator shall be tested for comparison with the prime standard in accordance with MIL-L-25467 (see 3.13.3).

4.6.2.1 Prime standard. Using parts which are identical to the parts to be used in production indicators, an indicator shall be constructed which meets all display requirements of MIL-L-25467. The indicator model shall be submitted to the procuring activity.

4.6.2.2 Prime standard life. A minimum of 12 average areas covering all representative areas of the face of the prime standard shall be measured for brightness and color prior to submission, and not less than every 50 hours of operation thereafter. A change of ±20% brightness or ±5% color change in any area using the initial data as a reference voids the prime standard certification.

4.6.2.3 Production (visual comparison). With 2.50 ±0.01 V simultaneously applied to the indicator under test and to the prime standard, like areas of the warning flag, moving background, numerals, indicia and markings shall be compared. Overall uniformity, similarity and brightness of the test indicator to that of the prime standard shall be observed. Like comparison of the test indicator to that of the prime standard shall be observed. Like comparison of the test indicator to that of the prime standard shall allow for acceptance. If the brightness is dissimilar in any individual area, 2.40 and 2.60 volts shall be applied consecutively to the prime standard. The brightness of the production indicator at 2.50 ±0.1 V shall fall within the brightness of the prime standard when set at 2.40 and 2.60 volts. A color comparison test shall be made with the prime standard and production indicator energized with 5.0

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±0.1 V. The indicators shall be identical in color. If any questionable areas still exist, they shall be measured with a photometer in accordance with the requirements of MIL-L-25467. If the production indicator fails this test, it shall be rejected and returned for rework.

4.6.3 Synchro and resolver zero. Each control transformer receiver synchro and resolver shall be checked for zero by the method specified in section 3 or an equivalent method acceptable to the procuring activity (see 3.15).

4.6.4 Azimuth servo operation. The indicator shall be properly connected. The rotor of the standard synchro test transmitter connected to the azimuth card servo shall be rotated at constant speeds of 15, 30, 45 and 60°/sec in both directions. The indicated heading shall not depart from the standard synchro test transmitter heading by more than 10° and the oscillation of the azimuth ring while tracking shall not exceed .5°. The heading and course pointers shall follow with the azimuth card (see 3.13.11).

4.6.5 Pointer servo operation. The standard synchro test transmitter shall be connected to each of the other servo systems and rotated at speeds of 15, 30, 45 and 60°/sec. The indicated pointer positions shall not depart from the standard synchro test transmitter by more than 10° and the oscillation of the pointer positions while tracking shall be not greater than .5° (see 3.13.26).

4.6.6 Pointers and dial sensitivity. The indicator shall be properly connected and the rotor of the standard synchro test transmitter shall be rotated one degree and there shall be a perceptible movement in each of the pointers and dial (see 3.16).

4.6.7 Scale error. The indicator shall be properly connected. The standard synchro test transmitter shall be set to zero and the heading indication shall be read and recorded. The standard synchro test transmitter shall be connected to the azimuth card and rotated successively to headings 30, 60, 90, 120, 150, 180, 210, 240, 270, 300, 330 and 360° and the readings under the lubber line shall be recorded. Any one reading shall not differ from the corresponding synchro heading by more than one degree. The identical procedure shall be followed for the heading and course marker except that the reading shall be with respect to the azimuth card. The identical procedure shall be followed for the bearing pointer except that the zero shall be at 180° (lubber line at the bottom) and the reading taken relative to this zero (see 3.17).

4.6.8 Servo follow-up rate and damping. The indicator shall be properly connected. The standard synchro test transmitter connected to the azimuth card shall be de-energized, its rotor rotated 90°, and re-energized. The time required for the azimuth ring to rotate between 5 and 85° of its original displacement from the standard synchro test transmitter shall be not greater than 4.5 seconds, and the total time before the azimuth ring comes to rest shall not exceed 5.5 seconds. The identical test shall be performed on each of the other servo systems (see 3.18).

4.6.9 Course bar mechanism. The course bar mechanism shall meet the dynamic response characteristics specified in 3.13.30.

4.6.10 Balance. A static balance test with up to 1 g applied to each axis of the case shall not cause the bar to deviate from its center position by more than .25 bar width (see 3.13.27).

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4.6.11 Accuracy. An accuracy test with ±DC applied to the proper terminals shall result in 75 ±6 microamperes/dot displacement (see 3.13.30).

4.6.12 Pointer freedom. Twice normal current shall be applied in each direction and slowly reduced. The pointer should not stick (see 3.19).

4.6.13 Flag alarm. The operating current shall be not greater than 0.5 milliampere (see 3.13.31).

4.6.14 To-from arrow. The sensitivity of the to-from arrow shall be checked by applying current in each direction as given in 3.14.32. Double voltage shall be applied in both directions and slowly reduced. There shall be no sticking (see 3.13.32).

4.6.15 Distance display. The standard synchro test transmitter shall be applied to each receiver synchro. A position test at every 36° shall be taken as specified in 3.13.36.

4.6.16 Power failure warning. The warning indicator shall meet the requirements of 3.13.28.

4.6.17 Shutter. Proper polarity of 28 V DC shall be applied to the shutter mechanism. The shutter shall disappear from view (see 3.13.38).

4.6.18 Case leakage. Each case shall be tested with a mass spectrometer or equivalent. The rate of leakage of the gas or mixture of gases in the case of the indicator proper, adjusted to a pressure differential of one atmosphere shall not exceed a rate which would permit more than 10 percent loss of the total filling medium after 1000 hours (see 3.4.4.2).

4.6.19 Lighting (visual comparison). The lighting circuit shall be tested by visual comparison of the production indicator against the prime standard. The tests shall be conducted as follows (see 3.13.3):

a. Color. Five volts shall be applied to both the indicator under test and the prime standard and the two shall appear identical in color.

b. Brightness. The prime standard and the production indicator under test shall be placed side by side at eye level. The two indicators shall be viewed at a distance of approximately two feet from the point within a cone 30° from the normal to the surface of the cover glass. The overall lighting of the two units shall be compared with 3.00 ±0.03 V applied to the lighting terminals of the indicator under test. General overall lighting level comparison shall then be made between the indicator being tested and the prime standard with 2.80 V applied and then varied up to 3.20 V. If the general overall lighting level of the indicator being tested does not fall within the brightness limits of the prime standard when energized at the specified voltage levels, the indicator shall be rejected. Indicators rejected due to this test shall be subjected to all display tests specified in MIL-L-25467.

4.6.19.1 ANVIS compatible lighting. When required as specified in the contract or purchase order, aviator's night vision imaging system compatible lighting inspection requirements shall be in accordance with MIL-L-85762 (see 3.13.1).

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4.6.20 Manufacturing run-in test. Each indicator shall be subjected to the conditions specified herein for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the indicator. Only those adjustments will be permitted which can be made by using such controls and adjustments are accessible to the operator during normal use of the indicator. The indicator shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the six hour period of operation. Where feasible, the indicator shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the indicator in two directions, the vertical direction shall be used. During the six hour period of operation following the 10 minute vibration period, the indicator shall be mechanically cycled periodically through its various phases of operation. Should failure occur, it shall be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future indicators. A record shall be kept of all failures. The six hour period specified above may be composed of two three hour periods to conform with standard working hours (see 3.20).

4.6.21 Power consumption. The power consumption of the indicator shall be measured with suitable meters. The alternating current consumption excluding lighting shall not exceed 55 VA (see 3.10).

4.6.22 Voltage and frequency variation. The indicator shall be properly connected with power applied and subjected to the transient power conditions specified in MIL-STD-704 for a total of 10 cycles at a rate of one hertz. No damage shall result. The indicator shall then be subjected to the individual tests at voltages and frequencies of 103V 380 Hz, 103V 420 Hz, 127V 380 Hz and 127V 420 Hz. The zero indication of the servoed pointers and azimuth card shall not shift more than one degree from the normal null position (see 3.10).

4.6.23 Low temperature operation. The indicator shall be properly connected, except that no power shall be applied and placed in a chamber at a temperature of -54°C $\pm$2° for a period of four hours. At the end of this period, and while the indicator is at the reduced temperature, power shall be applied for a period of one minute. At the end of this one minute period but still at this reduced temperature, the indicator shall meet the requirements of the individual tests (see 4.4.1 and 3.21).

4.6.24 High temperature operation. The indicator shall be properly connected and power shall be applied. The indicator shall then be subjected to an ambient temperature of 71°C $\pm$2° for a period of four hours. While at this temperature the indicator shall be subjected to the requirements of the individual tests (see 4.4.1). This test may be combined with the high temperature exposure test at the discretion of the testing activity (see 3.22).

4.6.25 Fogging. The hermetically sealed indicator shall be energized and placed in a 71°C $\pm$2° ambient temperature for a period of one hour. After this period and while still at this temperature, an ice cube shall be rubbed on the cover glass for a period of one to two minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence

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of water or oil fog inside the cover glass. Evidence of fogging shall be cause for rejection (see 3.23).

4.6.26 Magnetic effect. The indicator shall be properly connected and power applied. The indicator shall be revolved about a short bar magnetic compass with the nearest part of the indicator 12.0 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpindicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270 and 315°. At each of these positions the indicator shall be rotated 360° about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed one degree. This test shall be repeated with no power applied. The same tolerance shall apply (see 3.24).

4.6.27 Acceleration sampling plan A test. The indicator shall be properly connected and power applied to the servos and meter movements. The indicator shall be accelerated along each axis with a force of 5g. The servoed pointers and azimuth card shall not vary by more than one degree from the initial indication and the deviation bar shall not vary by more than one bar width from its initial position. The to-from arrow with no power applied shall not appear as a result of this acceleration (see 3.25).

4.6.28 Dielectric strength. A potential of 250V, 60 Hz AC shall be applied between isolated pins and between pins and the case for a period of five seconds. There shall be no breakdown of insulation or any other permanent damage to the indicator as a result of this test (see 3.26).

4.6.29 Vibration functional. The indicator shall be connected to the test transmitter and subjected to a functional vibration test in accordance with MIL-STD-810, Method 514.3, Equipment Category 5, Table 514.3-III, Figure 514.3-26, except that W_o shall be 0.02 g/Hz and the low frequency power spectral density shall be factored by 4 db/octave to 0.01 g/Hz. The indicator shall meet the requirements listed below and the requirement of the individual tests following the vibration test (see 3.27).

a. The servoed pointers shall not vary by more than one degree.

b. The azimuth card shall not vary by more than one degree.

c. The deviation bar shall not vary by more than one bar width.

d. The to-from arrow with no power applied shall not appear in its window.

4.6.30 Low voltage. The indicator shall be properly connected and the AC voltage shall be reduced to 103V, 380 Hz. The indicator shall meet the requirements of the individual tests (see 4.4.1 and 3.10).

4.6.31 Electromagnetic interference. The electrical system of the indicator shall be tested for conducted and radiated electromagnetic interference in accordance with MIL-STD-461 and MIL-STD-462 (see 3.6).

4.6.32 High temperature exposure. The indicator shall be properly connected and subjected to a high temperature exposure test in accordance with

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MIL-STD-810, Method 501.2, Procedure II, except the exposure time shall be 24 hours and no power shall be applied. At the end of the 24 hour period and while still at the high temperature, the indicator shall meet the requirements of the individual test (see 4.4.1). The indicator shall then be returned to standard conditions and allowed to stabilize for a period of four hours, after which it shall be subjected to and shall meet the requirements of the individual tests (see 3.22).

4.6.33 High altitude, low temperature. The indicator shall be subjected to the low temperature test in accordance with MIL-STD-810, Method 500.2, Procedures II and III, except that the examination shall be performed at the end of four hours (temperature at -54°C and pressure equivalent of 50,000 ft.). The time required for the indicator to start operating shall not exceed five minutes. While at -54°C and 50,000 feet the indicator shall meet the requirements of the individual tests (see 4.4.1). The indicator shall then be returned to the standard conditions and allowed to stabilize for a period of four hours, after which the indicator shall be subjected to and meet the requirements of the individual tests (see 3.28).

4.6.34 Temperature shock. The indicator shall be tested in accordance with MIL-STD-810, Method 503.2, temperature-shock test using a high temperature of 71°C ±2° and a low temperature of -54°C ±2°. After completion of the test procedure the indicator shall be subjected to and meet the requirements of the individual tests (see 4.4.1 and 3.29).

4.6.35 Acceleration sampling plan B test. The indicator not operating shall be mounted in its normal operating position and accelerated first along its vertical axis and then along each of two axes that are perpindicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20 g's in each of the above axes for a period of one minute. At the end of this acceleration, the indicator shall meet the requirements of the individual tests (see 4.4.1). No damage to the indicator shall result from this test (see 3.25).

4.6.36 Vibration endurance. The indicator shall be rigidly attached to a vibration fixture and vibration machine using the mounting holes. With no power applied the indicator shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.3, Equipment Category 5, Table 514.3-III, Figure 514.3-26, Curve A except that W_o shall be 0.04 g/Hz and low frequency power spectral density shall be factored by 4 db/octave to 0.02 g/Hz. At the conclusion of this vibration test, the indicator shall be subjected to and meet the individual tests. There shall be no damage to any part of the indicator as a result of this test (see 3.27).

4.6.37 Impact shock. The shock test shall be in accordance with MIL-STD-810, Method 516.3, Procedures 1 and 5 and as specified herein. The indicator shall be rigidly attached to the shock machine using the mounting holes. The indicator shall be connected and energized using standard transmitters during the shock test. Three shocks in each direction shall be applied along the three mutually perpindicular axes of the indicator (total 18 shocks). The shock pulse shall be one-half sinusoidal with a peak value of 15 ±2g and a duration of 11 milliseconds. The indicator shall be subjected to and comply with the requirements of the individual tests following completion of the shock test. A crash safety test shall be performed to establish the integrity of the mounting. The indicator or dummy load shall be attached to its normal mounting holes. The indicator or dummy load shall be subjected to

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two shocks in each direction along the three mutually perpindicular axes (total 12 shocks). The shock pulse shall be one-half sinusoidal with peak value of 30 ±5g and a duration of 11 milliseconds. There shall be no failure of the mounting and the indicator or dummy load shall remain in place and not create a hazard. Bending and distortion shall be permitted (see 3.30).

4.6.38 Humidity. The indicator shall be tested in accordance with the humidity test of MIL-STD-810, Method 507.2. Upon completion of the test, the indicator shall meet the requirements of the individual tests. There shall be no evidence of corrosion or rusting of the internal or external parts of the indicator assembly which would adversely affect subsequent operation (see 3.31).

4.6.39 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.2 for a period of 50 hours. At the end of the 50 hour period, there shall be no damage to the indicator which would adversely affect the operation of the required indicator. The individual tests shall apply (see 3.32).

4.7 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing this test. The test shall be for 1000 hours duration and shall be conducted on equipments that have passed the the reliability assurance production acceptance phase. The test shall be conducted under the conditions specified in 4.7.2. The life test sample shall be selected by the Government inspector (see 3.4.3).

4.7.1 Quantity to be selected for life test. One per contract of 300 equipments or less shall be selected. For contracts in excess of 300 indicators, quantity shall be specified in the contract.

4.7.2 Conditions for the life test. The indicator shall operate for a period of 1000 hours. The azimuth card, heading, course markers, bearing pointer and the distance display shall be cycled at a rate of 20°/sec. The direction of rotation shall be reversed every two minutes. At the end of the 1000 hour period, the indicator shall be subjected to and meet the requirements of the individual tests.

4.7.3 Lamp life. The indicator shall be subjected to the life test in MIL-L-25467 and may be combined with the life test (see 4.7.2 and 3.4.3).

4.7.4 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the indicator temperature to stabilize. A daily visual observation shall be made for proper operation.

4.7.5 Performance test. At approximately 100-hour intervals, the indicator shall be subjected to and shall meet the requirements of the individual tests.

4.7.6 Test data. The contractor shall keep a daily record of the performance of the indicator, making particular note of the deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

a. Part type number

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b. The circuit reference symbol number

c. The part function

d. Name of manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to the failures.

4.7.7 Failure reports. In the event of a failure, the Government inspector shall be notified immediately. The contractor shall propose suitable and adequate design or material corrections for failures which occurred.

4.7.8 Fungus resistance. The indicator shall be tested in accordance with MIL-STD-810, Method 508.3. No deterioration that would adversely affect subsequent operation will be sustained (see 3.33).

4.7.9 Structural test. A test shall be conducted to ascertain that neither the indicator electrical connector nor the indicator case are damaged or deformed in any way when a total force of 60 lb is applied to the connector along its longitudinal axis, while inserting the indicator into a stationary mount. The testing apparatus and procedure to be utilized in conduct of this test shall be approved by the acquiring activity prior to conducting the test (see 3.35).

4.7.10 Maintainability demonstration. The maintainability demonstration test shall be conducted in accordance with MIL-STD-471. The tests shall be selected in advance and shall be approved by the procuring activity. The instrument shall have a median value of corrective maintenance tasks (Mct) of 6.0 hours when Test Method 1, Test Plan A of MIL-STD-471 is used. The maximum Mct shall be no longer than 8.0 hours. Maintainability demonstration shall be performed on a minimum of three selected repairs.

4.7.11 Criteria for failure. Deterioration or change in performance of any components which could in any manner prevent the indicator from meeting functional, maintenance and service requirements during service life shall provide reason to consider the indicator as having failed to comply with the conditions of the test to which it was subjected.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial (see 6.2).

5.1.1 Level A. The indicators shall be preserved in accordance with MIL-P-116, Method III.

5.1.1.1 Cleaning. The cleaning of the indicators shall be in accordance with MIL-P-116, Procedure D4.

5.1.1.2 Drying. The drying of the indicators shall be in accordance with MIL-P-116, Procedure D4.

5.1.1.3 Preservative. No preservative is required.

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5.1.1.4 Unit pack. One cleaned and dried indicator shall be placed in a container conforming to PPP-B-1672, Type I (refer to MIL-STD-2073/1A, Appendix E for additional information pertaining to the selection of the proper size container for the weight, cube and fragility factor of the indicator).

5.1.2 Commercial. The indicators shall be preserved in accordance with ASTM-D-3951.

5.2 Packing. Packing shall be level A, level B or commercial.

5.2.1 Level A. Indicators preserved as in 5.1.1 shall be packed in shipping containers conforming to PPP-B-585, Class 3 or MIL-B-43666, Type I or II or PPP-B-601, Type overseas. The number of unit packs per shipping container shall be as specified in the contract (see 6.2).

5.2.2 Level B. Indicators preserved as in 5.1.1 shall be packed in a shipping container conforming to PPP-B-636, weather-resistant, PPP-B-591, weather-resistant or PPP-B-640, weather-resistant. The number of unit packs per shipping container shall be as specified in the contract (see 6.2).

5.2.3 Commercial. Indicators shall be packed in accordance with ASTM-D-3951.

5.3 Marking. The interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking. The following precautionary marking shall appear conspicuously on two opposite sides of each interior and shipping container:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Associated equipment. The indicator covered by this specification is a component of an integrated flight instrumentation system conforming to MIL-S-23535. The other components are:

a. Flight director computer - MIL-C-23367

b. Remote attitude director indicator - MIL-I-23524

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. The quantity and part number of the instrument desired.

c. Levels of packaging (see 5.1) and packing (see 5.2) desired.

d. The laboratory that shall conduct tests (see 4.4, 4.4.2.2).

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e. Whether fungi tests of component parts or assembled indicator is to be accomplished (see 3.3.3.6).

f. Quantity to be selected for life test (see 4.7.1).

g. Whether or not ANVIS compatible lighting is required.

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Discription (DD FORM 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD FORM 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.
4.4, 4.7.2	Reports, Production Inspection	DI-T-4904
4.1.2, 4.4.3	Procedures, Test	UDI-21347

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12.L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Hermetic seal. Hermetic sealing is the process by which an item is totally enclosed by a suitable metal structure or case and sealed airtight by fusion of metallic or ceramic materials. This includes the fusion of metals by welding, brazing or soldering, the fusion of ceramic materials under heat or pressure and fusion of ceramic materials into metallic support. Elastomeric or resinous materials or combinations of these materials may be used as a cover-glass seal provided that the specific leak-rate requirements are met under all tests and environmental conditions listed herein. Elastomeric or resinous materials may be used in other areas provided prior approval is obtained from the procuring acitivity.

6.3.2 Good as new. "Good as new" shall mean indicators operated less than 10% of the specified MTBF operation.

6.3.3 Refurbished. Refurbished shall mean that the indicator has been completely overhauled with all component parts meeting current parts standards, and the indicator shall have been subjected to, and meet, all the requirements for a new indicator.

6.3.4 Indicator. Unless otherwise specified in a particular test, the word "indicator" as used in section 4 of this specification shall be construed to mean the ID-1013( )/A indicator.

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6.4 First article. When first article inspection is required, the indicator will be tested and should be a first article sample. The first article should consist of the three samples as specified in 4.3.1. The contracting officer should include specific instructions in acquisition documents regarding examinations, tests and approval of first article.

6.4.1 First article provision. The manufacturer of indicators on contract should not commence until the samples submitted are pronounced satisfactory by the acquiring activity. When a contractor is in continuous production of the indicator from contract to contract, the submission of further first article samples on the subsequent contracts may be waived at the discretion of the acquiring activity. Approval of first article samples or the waiving of first article tests does not eliminate the requirement of quality conformance inspection.

6.4.2 First article information. It shall be understood that the indicator supplied under contract should be identical to the corresponding first article sample in material design, construction, quality, workmanship and method of manufacture. Deviation from the standards of first article sample should be made only by the acquiring activity. Evidence of unauthorized change should constitute cause for rejection.

6.5 Precedence of documents. When the requirements of the contract, this specification or applicable subsidiary specifications conflict, the following precedence should apply:

a. Contract. The contract should have precedence over any specification.

b. This specification. This specification should have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, should be specifically approved in writing by the acquiring activity.

c. Referenced specifications. Any referenced specification should have precedence over all applicable subsidiary specifications referenced herein.

6.6 Changes from previous issue. Astericks are not used in this revision to identify changes with respect to previous issues due to the extensiveness of the changes.

6.7 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement STANAG No. 3741 and AIR STD 10/45. When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take the appropriate action through the international standardization channels including department standardization offices to change the agreement or make other appropriate accomodations.

6.8 Subject term (key word) listing.

Indicator, horizontal situation
Aircraft
Instrument

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Hermetic sealing

Custodians:
Navy - AS
Army - AV

Preparing activity:
Navy - AS

(Project No.6610-0124)

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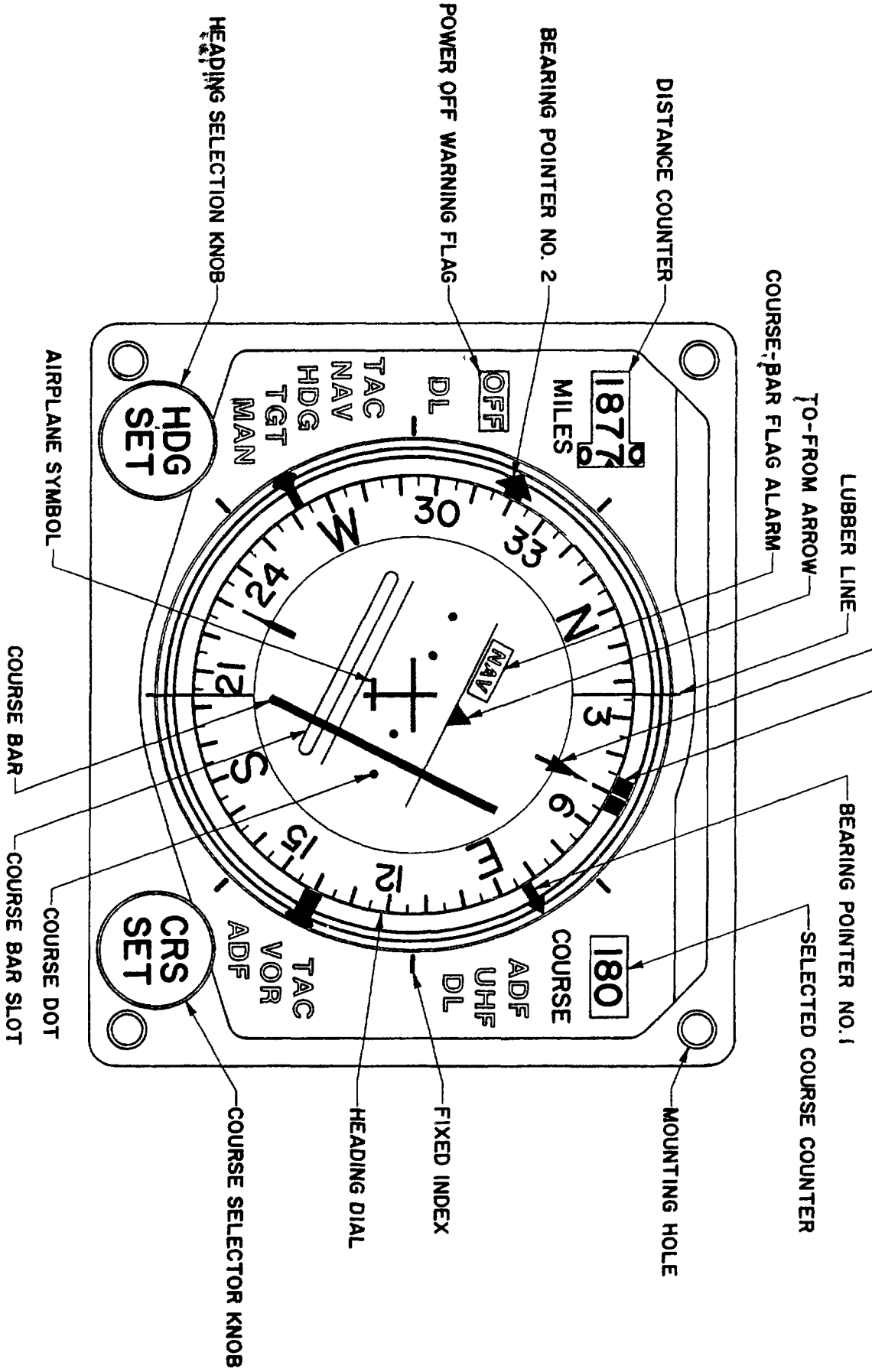
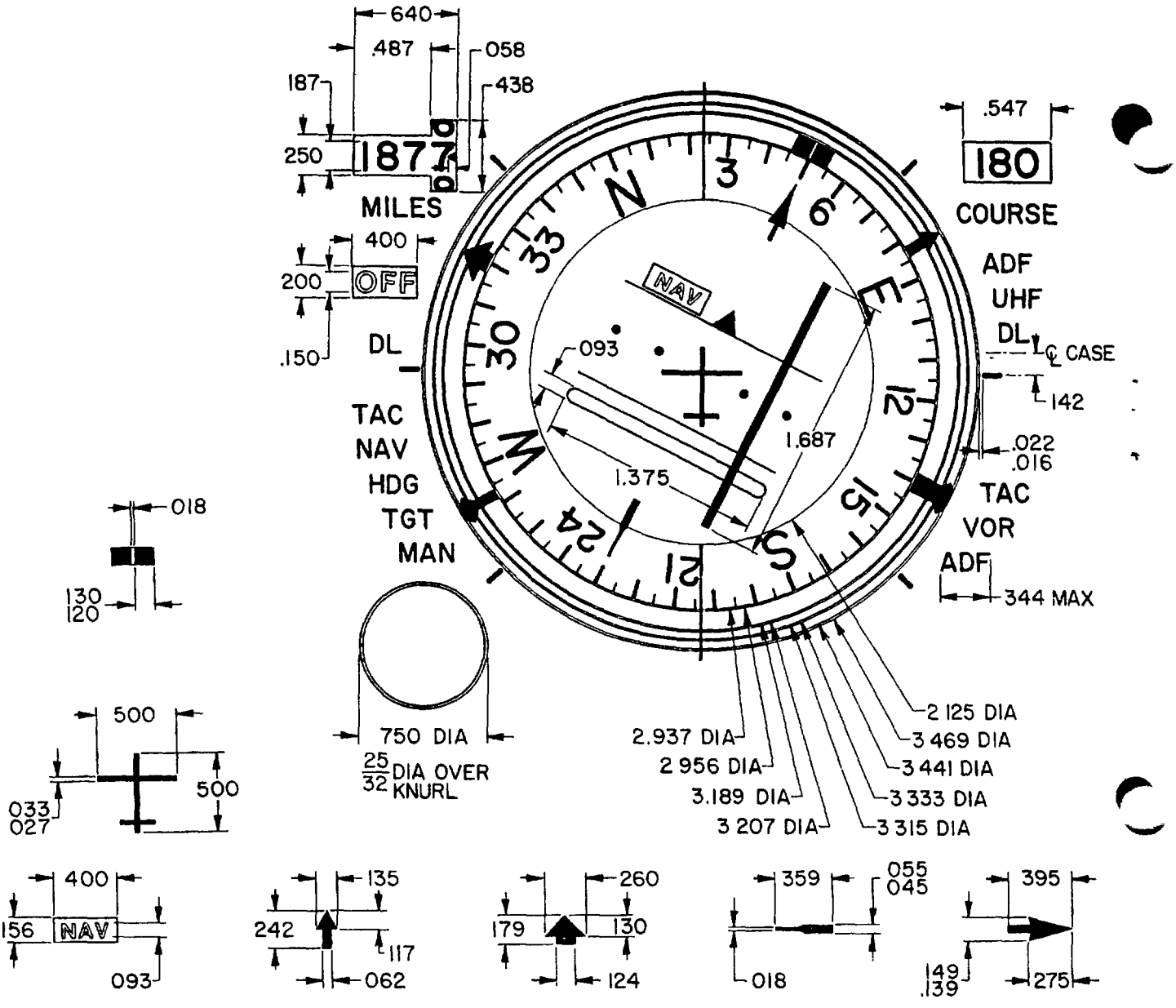


FIGURE 1. Indicator face design.

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MARKING	HEIGHT OR LENGTH ±0.001	WIDTH OF LINE ±0.002	MATERIAL OF FINISH
NUMERALS 3, 6, 12, 15, 21, 24, 30 AND 33	0.175	-----	
N, E, S AND W	0.218	-----	
1877, 180	0.187	-----	
MILES, DL, TAC, NAV, HDG, TGT, MAN COURSE, ADF, UHF, DL, TAC, VOR AND ADF	0.125	-----	
MAJOR GRADUATIONS	0.125	0.020	Lusterless white. Color No. 37873 of FED-STD-595
MINOR GRADUATIONS	0.062	0.010	
COURSE DOTS	0.052 DIA	-----	
LUBBER LINE	-----	0.015	
FIXED INDEX BARS	0.125	0.028	
EQUILATERAL TRIANGLE	0.156	-----	
AIRPLANE SYMBOL	-----	-----	
COURSE BAR	-----	-----	
POINTERS	-----	-----	
BACKGROUND OF OFF AND NAV	-----	-----	
OFF	0.150	-----	Lusterless black Color No 37038 of FED-STD-595
NAV	0.098	-----	
BACKGROUND OF DIAL	-----	-----	

FIGURE 2. Indicator face dimensions.

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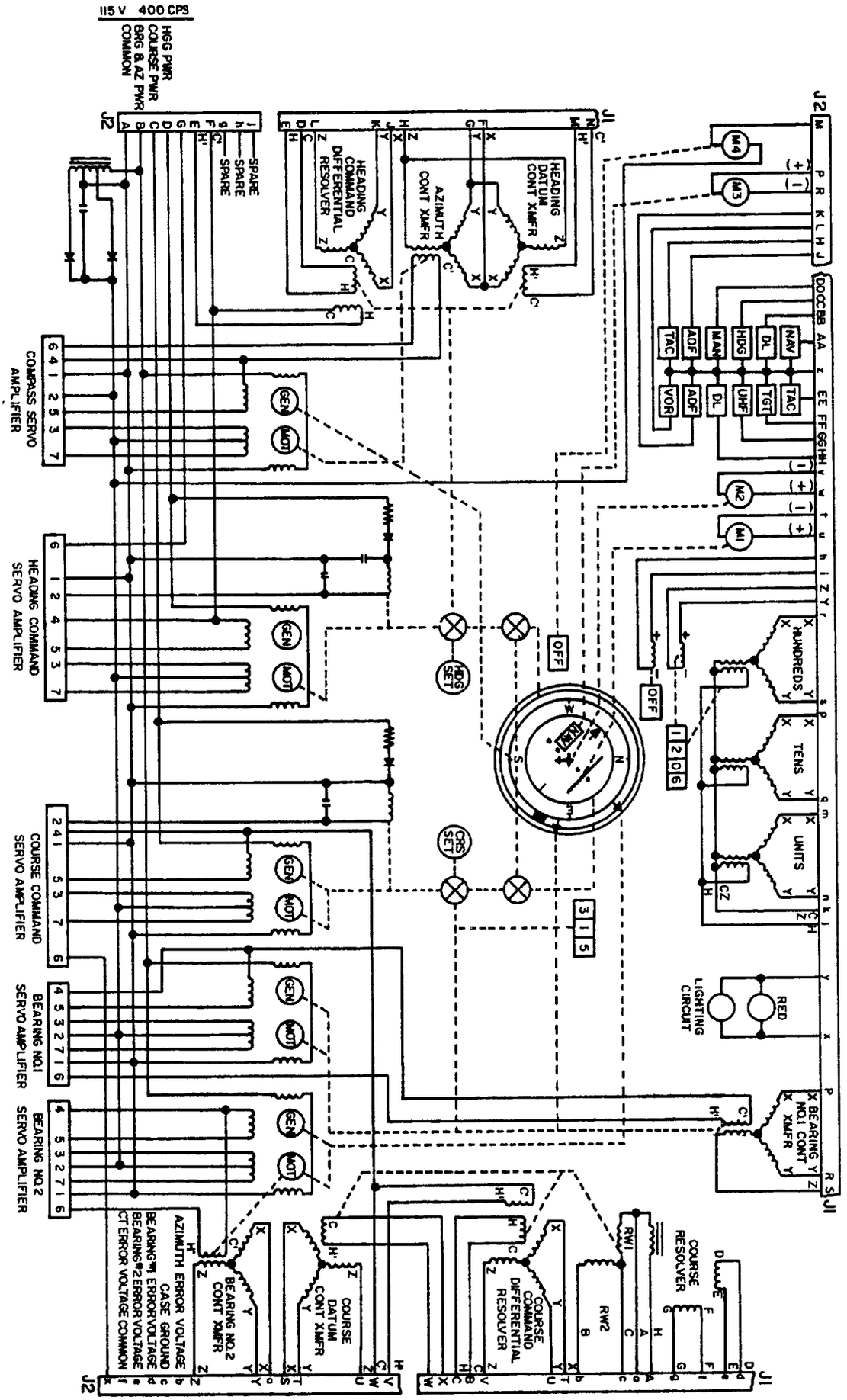


FIGURE 3. Schematic wiring diagram.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-23366B	Indicator, Horizontal, Situation ID-1013

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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